

E1:

This device uses several key STEM principles to function such as air pressure, chemical properties and mathematical equations to figure out pressure and convert units. We are using a valve and gauge system to measure the air pressure exhaled by a person. Once the air travels down the airtight tube it reaches the gauge and it measures the resistance on its internal parts and outputs the information regarding the density of air molecules in the tube. The chemical properties of the device include plastic tubing, a non toxic, 3D printed tube that will attach to the mouthpiece to create accurate readings.

Mathematical Equation to covert from mmHg to cmH₂O

To convert pressure from millimeters of mercury (mmHg) to centimeters of water (cmH₂O):

$$\text{Pressure (cmH}_2\text{O)} = \text{Pressure (mmHg)} \times 1.35951$$

Explanation:

- 1 mmHg = 1.35951 cmH₂O (based on the densities of mercury and water).
- Multiply the value in mmHg by this conversion factor to get the equivalent in cmH₂O.

Equations to calculate the Spring Constant

While brainstorming designs, we decided to pursue a potential spring prototype. In order to have a spring that is able to compress due to air pressure, we had to calculate the spring constant. To calculate it:

$$\text{Force} = -kx$$

Explanation:

- k stands for the spring constant
- x stands for the distance

Experiment:

A spring constant experiment typically involves measuring how a spring stretches or compresses under an applied force. The goal is to determine the spring constant (k) using Hooke's Law, which states:

Where:

- F is the applied force (N),
- k is the spring constant (N/m),
- x is the extension or compression (m).

Procedure for Measuring the Spring Constant

Materials Needed:

- A helical spring

- A set of known weights (e.g., 100 g, 200 g, etc.)
- A meter ruler
- A clamp stand with a hook
- A pointer (optional, to improve measurement accuracy)

Steps:

- Set Up the Apparatus: Attach the spring to the clamp stand and ensure it hangs freely.
Place the ruler vertically beside the spring.
- Measure the Initial Length: Record the unstretched length of the spring.
- Apply a Known Mass: Hang a weight on the spring and allow it to come to rest. Record the new length.
- Calculate Extension: Compute the extension using $x = L - L_0$.
- Repeat for Different Masses: Increase the mass incrementally and record the corresponding extensions.
- Calculate the Force: Convert the mass to force using $F = mg$ (where $g = 9.8 \text{ m/s}^2$).
- Graph the Results: Plot a graph of force (F) vs. extension (x). The slope of the best-fit line gives the spring constant (k).

Analysis & Interpretation

- A linear relationship confirms Hooke's Law.
- The slope of the F-x graph gives , meaning:

- If the force exceeds the elastic limit, the spring will not return to its original length, indicating permanent deformation.

Application of STEM Principles and Practices to Design Requirements

Durability/Longevity

- We utilized materials science principles to select components that ensure long-term durability and resistance to environmental stress. For the outer casing, we are considering high-strength, impact-resistant plastics or lightweight aluminum, both chosen for their proven tensile strength, wear resistance, and ease of sanitation. Through stress analysis and fatigue testing, we will validate the device's reliability, ensuring it maintains precision and functionality for a minimum of five years while meeting our design requirements for durability, ease of use, and cost-effectiveness.

Accuracy

- Accuracy is one of the most critical aspects of our design. To ensure an accuracy of ± 1 cm H₂O, we applied principles from pressure systems. Using a diaphragm or differential pressure sensor, we can precisely measure the user's inspiratory pressure. We also integrated calibration techniques based on known medical standards to maintain consistent measurement accuracy. Mathematical practices, such as error analysis, will be used to identify and reduce potential sources of error in the readings.

Simplicity and Replicability

- Given the need to produce 10-20 devices, we focused on a design that minimizes complexity while maintaining functionality. Advice from our teachers and primary stakeholders guided us in developing a modular design that is easy to assemble and replicate. The simplicity of the design ensures that college-level students can understand and use the device without extensive instruction.

Affordability

- Affordability is crucial for our project, given the budget constraints of a high school engineering class.

By applying cost analysis and supply chain management principles, we selected inexpensive yet reliable components. This ensures that the device can be produced in large quantities without exceeding the budget. Additionally, using open-source design elements reduces the need for costly proprietary parts.

Portability

- We used OnShape and other CAD softwares to design our device to minimize the size of the device and calculate the weight and minimize volume. This allows for the device to be more easily transported and for students to trade them off with one another with ease.

E2.

Our solution is well-supported by STEM principles. Our solution may require a change between the units of measurement (from mmHg to cmH₂O, adhering to the students' [of the Drexel class] needs), which required some mathematical calculations. Additionally, while pursuing a spring prototype, we had to perform calculations/an experiment to determine the appropriate spring constant, ensuring the prototype responds effectively and accurately to the change in air pressure. The prototype adheres to our criteria, ensuring that we and the Drexel students can have a working device to aid the .

Material Selection for Durability:

- Using high-strength plastics or aluminum ensures the device withstands frequent handling. These materials were selected after applying **mechanical properties analysis**, such as tensile strength and impact resistance.

Sensor Selection for Accuracy:

- The pressure sensor was chosen based on **pressure calibration standards**. By comparing known air pressures to our device, we are able to form an **error analysis** and can ensure that the device remains accurate over time.

Ease of Assembly for Simplicity and Replicability:

- We followed **modular design principles** to simplify the assembly process. Each module can be independently produced and tested, allowing for replication of the units the class needs.

Cost Optimization for Affordability:

- Our approach involved conducting a **cost-benefit analysis** of various components. By sourcing parts from low cost, yet reliable sources, such as 3d-printing and trustworthy stores, we ensured the overall production cost remains within budget.

Compact Design for Portability:

- We applied **Ergonomics and human factors engineering** to make the device easy to carry and store. The compact design not only aids portability but also protects the device from damage during transport.

E3. (&E4.&E5.)

This design was checked by Dr. Joseph Sarver and Dr. Jaimie Dougherty. Dr. Joseph “J” Sarver is a faculty member at the Drexel School of Biomedical Engineering, Science, and Health Systems. He has studied the intervertebral discs of mice as an animal model to study degeneration. Additionally, during the ‘REMO’ project, he helped research remobilization following a rotary cuff repair in an animal model and non-contact intra-tendinous strain using texture tracking with an MRI. More recently, he presented at the 20th European meeting on supercritical devices on research done in partnership with PureCycle Technologies.

Dr. Jaimie “B” Dougherty is also a faculty member at the Drexel University’s School of Biomedical Engineering, Science, and Health Systems. She has a PhD in Biomedical Engineering with a focus on fatigue analysis using electromyography in the MR environment, so she has the expertise in analyzing functional outputs and biomedical device application. Due to her work, she has been awarded a postdoctoral fellowship to study cortical encoding in spinal cord injury recovery at Shriners Hospital for Children. She has also done work in brain-computer interface (like a neurolink) and electrophysiological signal acquisition and processing.

Both experts have documented relevant credentials in fields directly related to the project. Their knowledge in biomechanics, device design, and functional measurement systems qualifies them to provide feedback on the STEM principles and design elements of our project. They are highly skilled in their field and their insight is valued very much.

Expert Feedback from the presentations:

- **Calibration:**
 - The experts told us that in terms of calibration, we should aim for accuracy, but it doesn't need to be perfect. They wanted us to make sure that, when checking the calibration, we use a known reference point/device and it should be as close as possible to the real pressure.
- **Durability:**
 - We want this to be able to last about 5 years, so both experts highlighted the importance of minimizing the number of moving or fragile parts to improve durability.
- **Learning Objectives:**
 - The experts emphasized that the goal of this project is not only to provide accurate measurements but also to help students learn from the process. They suggested that if a digital display is included, it should serve as a tool to help students interpret the data easily.

In our prototype, we tried to adhere to these suggestions by limiting the amount of parts we have in the design. So, we only have a dial piece, tube, screw to connect the tube and dial, and mouthpiece. Additionally, we bought two dials to compare (when we tweak one of them, we can double check the calibration with the other) and are following up with Dr. Brossman to make sure we are still following her expectations.

Email log:

November 19th, 2024

Hello Team,

This is well done. I would offer that portability is not a huge issue as we can have an identified place for this device to sit (on a counter depth counter or on a mobility cart) if it is a larger device. The students can "move to it". It would have to be able to be portable in the sense that it can get from your HS to Drexel University (fit in an elevator, car trunk ect.).

I am still working on a budget- but I don't expect a large dollar amount. This I will keep working on!

Dr. Brossman

Heather Brossman PT, DPT,DHSc

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*Please forgive the typos as this is send from my phone *